

Multiple-Angle Identities

Date _____ Period _____

Use the half-angle identities to find the exact value of each.

1) $\sin 292.5^\circ$

2) $\sin 337.5^\circ$

3) $\tan 105^\circ$

4) $\cos 105^\circ$

5) $\cos 157.5^\circ$

6) $\cos 337.5^\circ$

Find the exact value of each.

7) $\tan \theta = \frac{15}{8}$ where $180 \leq \theta < 270$

Find $\tan 2\theta$

8) $\cos \theta = -\frac{2\sqrt{14}}{9}$ where $90 \leq \theta < 180$

Find $\sin 2\theta$

9) $\tan \theta = \frac{\sqrt{2}}{4}$ where $0 \leq \theta < 90$

Find $\cos \frac{\theta}{2}$

10) $\tan \theta = \frac{\sqrt{3}}{3}$ where $0 \leq \theta < 90$

Find $\cos 2\theta$

11) $\cos \theta = \frac{3}{5}$ where $270 \leq \theta < 360$

Find $\cos 2\theta$

12) $\tan \theta = \frac{4}{3}$ where $180 \leq \theta < 270$

Find $\tan \frac{\theta}{2}$